

# Luca Atella

Software Architect · Backend & Platform Engineering · AWS Certified  
Basilicata, Italy

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## Summary

Software Architect, Tech Lead, and Backend/Platform Engineer focused on **modular systems, cloud-portable architectures, runtime governance, and long-lived backend platforms**.  
I design and directly build backends, APIs, and application infrastructure. At Vemar, from April 2021 to October 2024, I worked as Tech Lead with continuous technical coordination responsibility for a 4-person team, personally designed and implemented the infrastructure for 80 trackers, and developed the framework on which delis.app is based.

## Certifications

- AWS Certified Solutions Architect - Associate
- AWS Certified Cloud Practitioner
- AWS Certified AI Practitioner
- Cisco Networking Academy - CCNA path

## Core Strengths

- Backend and platform architecture for integration-heavy systems
- AWS serverless deployment and managed-service architectures
- Cloud-portable application design and infrastructure abstraction
- Modular systems, plugin-based architectures, and runtime governance
- Docker / Docker Compose local environments and deployment automation
- IoT, edge/cloud integration, and Digital Twin architecture concepts
- Geospatial data processing and 3D terrain model generation

## Professional Experience

### Tech Lead / Backend & Platform Engineer — Vemar S.A.S.

April 2021 - October 2024

- Continuously coordinated a 4-person technical team
- Developed the plugin-based backend framework on which delis.app is based
- Personally designed and implemented the infrastructure for 80 trackers
- Worked on socket communication, CAN bus data, persistence, and REST API exposure
- Centralized application logic, authentication, API exposure, and persistence concerns
- Defined REST API contracts and backend structure
- Designed plugin loading and execution boundaries
- Introduced Docker-based development and deployment workflows

## Selected Projects

### Cloud-Portable Fleet Management Platform

- Fleet Management System used by Studio Lambda to manage vehicle turnover for 50 employees
- Built a FastAPI backend and worked on React/TypeScript frontend integration
- Abstracted persistence and object storage through repository and storage contracts
- Used Docker Compose, DynamoDB Local, and MinIO for local development
- Deployed to AWS using Lambda, ECR, DynamoDB, S3, CloudFront, CloudFormation, and CloudWatch

### Real-Time Fleet Telemetry Backend

- Infrastructure distinct from the Studio Lambda platform
- Personally designed and implemented at Vemar for 80 trackers
- Included socket communication, CAN bus data, persistence, and REST API exposure

### Unified Backend for Heterogeneous Data Sources

- Plugin-based backend framework I developed at Vemar and on which delis.app is based
- Integration separation, explicit contracts, and unified API model presented in SAFE form
- Does not attribute every architectural generalization to delis.app itself

**ImportSpy (Open Source)**

- Runtime contract enforcement engine for Python modules
- YAML-based declarative contracts for structural, contextual, and execution constraints
- Deterministic import-time validation model for modular and long-lived Python systems
- 15,000+ downloads on PyPI and presentation at GDG Basilicata

**B3DO — Basilicata 3D Open**

- Unpublished geospatial processing pipeline for transforming public Basilicata terrain datasets into 3D terrain models
- CLI-driven workflow for DTM merging, boundary extraction, raster clipping, resampling, LOD mesh generation, and textured previews
- Uses GDAL, Rasterio, NumPy, PyVista, Fiona, and Typer
- Separates raw data, processed datasets, build artifacts, and model outputs for reproducible data-to-model processing

**IoT Data Aggregation Architecture for Digital Twin Systems**

- Conceptual edge-to-cloud architecture for heterogeneous IoT data acquisition and aggregation
- Includes BLE/Zigbee mesh, gateway ingestion, GraphQL API, data-specific layers, microfrontends, storage gateway, and AI-assisted processing

**Runtime Compatibility for Agenzia delle Entrate Applications**

- Professional intervention in an accounting firm Windows environment
- Isolated Java runtimes for IRAP 2019-2021, ISA, Desktop Telematico / Entratel, and other tools
- Handled JNLP/LaunchAnywhere startup modes and Entratel TLS/truststore requirements
- Infrastructure and compatibility work, not tax consulting or regulatory certification

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**Education**

**BSc in Computer Science and Technology (L-31)** — University of Basilicata  
2018 - Present

**High School Diploma in Electronics, Electrical Engineering, and Automation**  
2013 - 2018

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**Languages**

Italian: native · English: B1, actively improving

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**Tech**

Python · FastAPI · React · AWS · Lambda · DynamoDB · S3 · CloudFront · CloudFormation · Docker · Docker Compose · Kubernetes · Linux · Networking · Runtime Governance · GDAL · Rasterio · PyVista